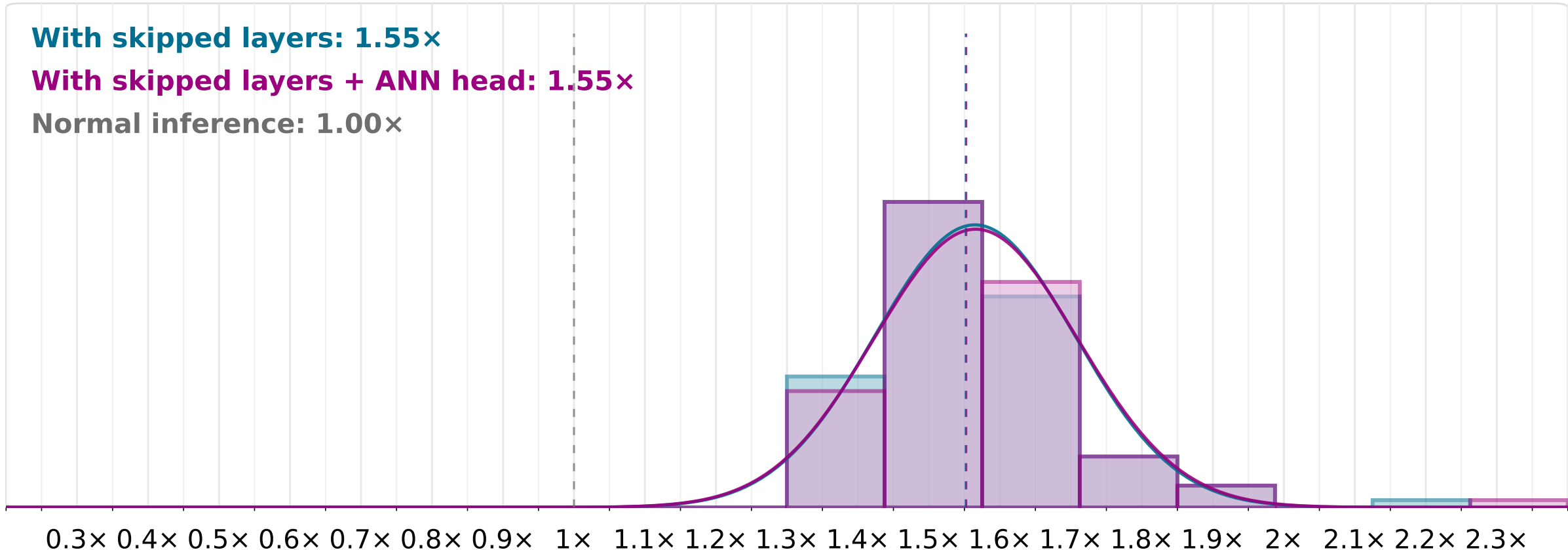


Self-speculation per-token speedup

Mistral 7B Instruct v0.3



Setup

Prompt set = **python diverse**
Gap = **(1, 1)**
Block size = **2**
Layers = **32**
LM vocab = **32,768**
LM params = **7.25B**
Head params = **134.2M**
Head portion = **1.85%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **100**

Results

Peak mem normal = **13.61 GiB**
Mean speedup (w/o ANNH) = **1.55×**
Acceptance rate (w/o ANNH) = **47.5%**
Exact match (w/o ANNH) = **58.0%**
Peak mem self (w/o ANNH) = **13.61 GiB**
Mean speedup (w ANNH) = **1.55×**
Acceptance rate (w ANNH) = **46.9%**
Exact match (w ANNH) = **58.0%**
Peak mem self (w ANNH) = **13.65 GiB**
Tokens gen = **s 19,509 / n 19,523**

Profile

Drafter split (w/o ANNH) = **B 74.2% / H 17.2% / O 8.7%**
Verifier/normal (w/o ANNH) = **1.06×**
Drafter/normal (w/o ANNH) = **10.0%**
Drafter split (w ANNH) = **B 83.8% / H 6.2% / O 10.0%**
Verifier/normal (w ANNH) = **1.06×**
Drafter/normal (w ANNH) = **8.9%**
ANNH accuracy = **98.7%**
ANNH head speedup = **3.09×**
ANNH index = **4096 clusters; top-k 100**